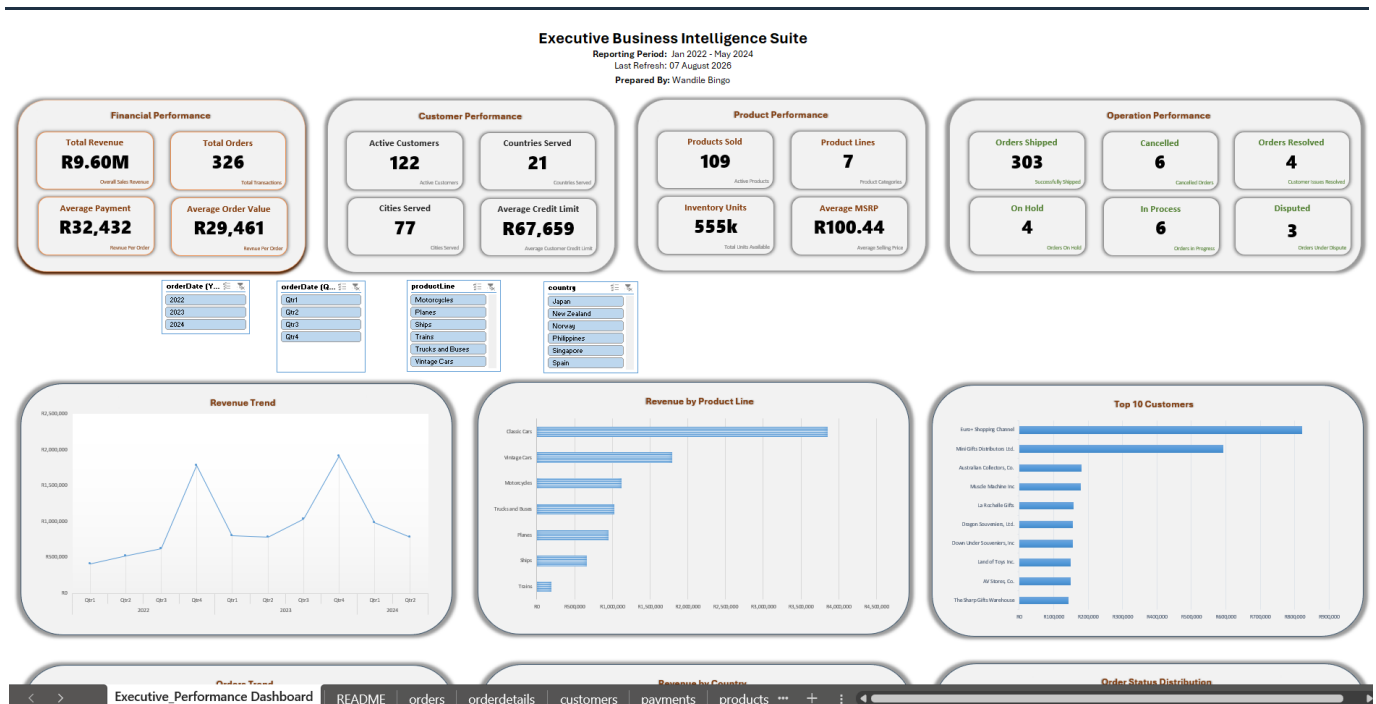


EXECUTIVE BUSINESS INTELLIGENCE SUITE

Business Operations & Performance Analytics | Executive Case Study

Microsoft Excel • Power Pivot • DAX • Relational Data Modelling • Executive Reporting



Executive Business Intelligence Suite — consolidated management dashboard

Prepared by Wandile Bingo | Reporting period: January 2022 – May 2024

Executive Brief

This case study presents an Excel-based executive management information solution that consolidates eight linked business datasets into a single decision-support environment. The solution was designed to give management a concise view of performance across revenue, customers, products and order fulfilment, while retaining the ability to interrogate the underlying drivers by period, market and product line.

REVENUE	ORDERS	CUSTOMERS	SHIPPED
R9.60m	326	122	92.9%
total sales	customer orders	customer master	of all orders

What the analysis says

- 2023 revenue was approximately R4.52m versus R3.32m in 2022 — an increase of about 36% across the two complete reporting years.
- Q4 is the clearest recurring demand peak: approximately R1.78m in Q4 2022 and R1.91m in Q4 2023, supported by 57 and 59 orders respectively.
- Classic Cars generated approximately R3.85m, representing about 40% of total revenue and creating both commercial strength and product-concentration exposure.
- The United States generated approximately R3.27m, or about 34% of total revenue; Spain and France form the next material revenue pools.
- The top 10 customers contributed approximately R2.68m, or 28% of revenue. Concentration is meaningful but not dominated by a single account.
- 303 of 326 orders were shipped. The 23 non-shipped orders are a small share of volume, but provide a focused exception set for operational review.

DECISION AGENDA

Prioritise Q3 readiness for the recurring Q4 demand peak; protect the Classic Cars franchise while developing adjacent categories; deepen high-value markets without increasing geographic dependence; formalise key-account management; and introduce recurring exception reporting for non-shipped orders.

Scope and reporting discipline

The dashboard is intentionally selective. It distinguishes transactional measures that can be filtered through the data model from master-data measures that should remain static where the source data does not contain historical snapshots. This avoids creating apparent interactivity at the expense of analytical accuracy. The 2024 period is also treated as partial-year data (January to May) and is therefore not presented as directly comparable with the complete 2022 and 2023 years.

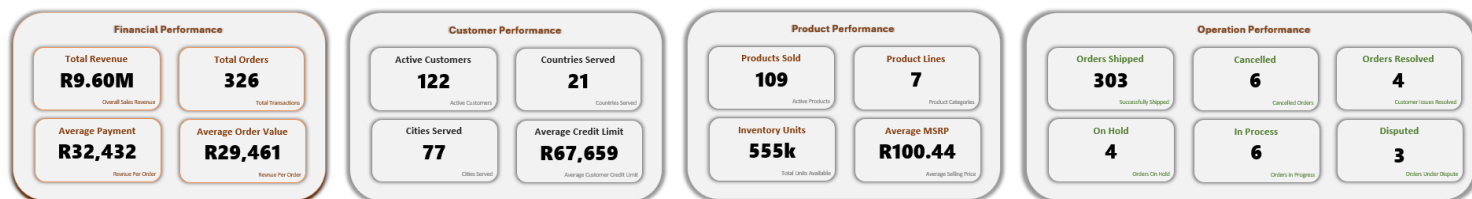
1. Business context and management objective

The source environment contains separate records for orders, order lines, customers, payments, products, product lines, employees and offices. In isolation, each table answers a narrow operational question. The management challenge is to connect those records into a coherent view of business performance without flattening the information into an unwieldy spreadsheet or forcing executives to interpret transaction-level detail.

The Executive Business Intelligence Suite was therefore built around a management objective rather than around a list of Excel features: provide a reliable, navigable view of performance that shows the scale of the business, the drivers of revenue, the concentration of value, the quality of order execution and the issues that warrant management action.

Management questions addressed

- How is revenue and order activity developing over time, and is there evidence of seasonality?
- Which product lines and geographic markets account for the largest share of value?
- How concentrated is revenue among the largest customers?
- What proportion of orders are successfully fulfilled and where do exceptions occur?
- Which measures should respond to executive filters, and which should remain static because the source data does not support historical filtering?
- What actions follow from the observed patterns?



Executive KPI layer — financial, customer, product and operational performance

2. Data architecture and analytical model

The model uses eight logically separate datasets and preserves the relationships between business entities. This design supports analysis across dimensions while avoiding unnecessary duplication of customer, product and organisational attributes.

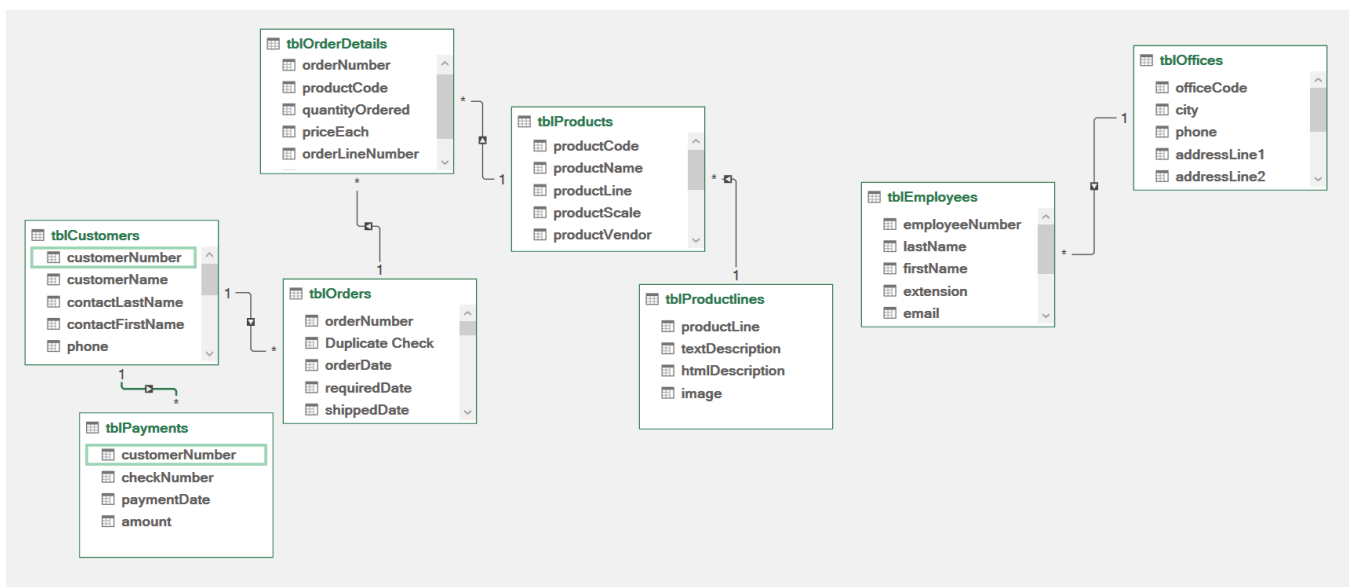
Dataset	Scale	Analytical role
Orders	326 orders	Dates, status, customer activity and order volume
Order Details	2,996 lines	Quantity, selling price and revenue at line level
Customers	122 customers	Customer identity, geography, sales representative and credit limit
Payments	273 payments	Customer payment values
Products	110 products	Product, inventory, pricing and product-line attributes

Product Lines	7 categories	Portfolio segmentation
Employees	23 employees	Sales-representative structure
Offices	7 offices	Organisational location and territory context

Revenue is derived at order-line level as Quantity Ordered × Price Each. The date hierarchy is retained at year, quarter and month level to support trend and seasonal analysis.

Relationship logic

- Customers → Orders via customerNumber
- Orders → Order Details via orderNumber
- Products → Order Details via productCode
- Product Lines → Products via productLine
- Customers → Payments via customerNumber
- Employees → Customers via salesRepEmployeeNumber
- Offices → Employees via officeCode



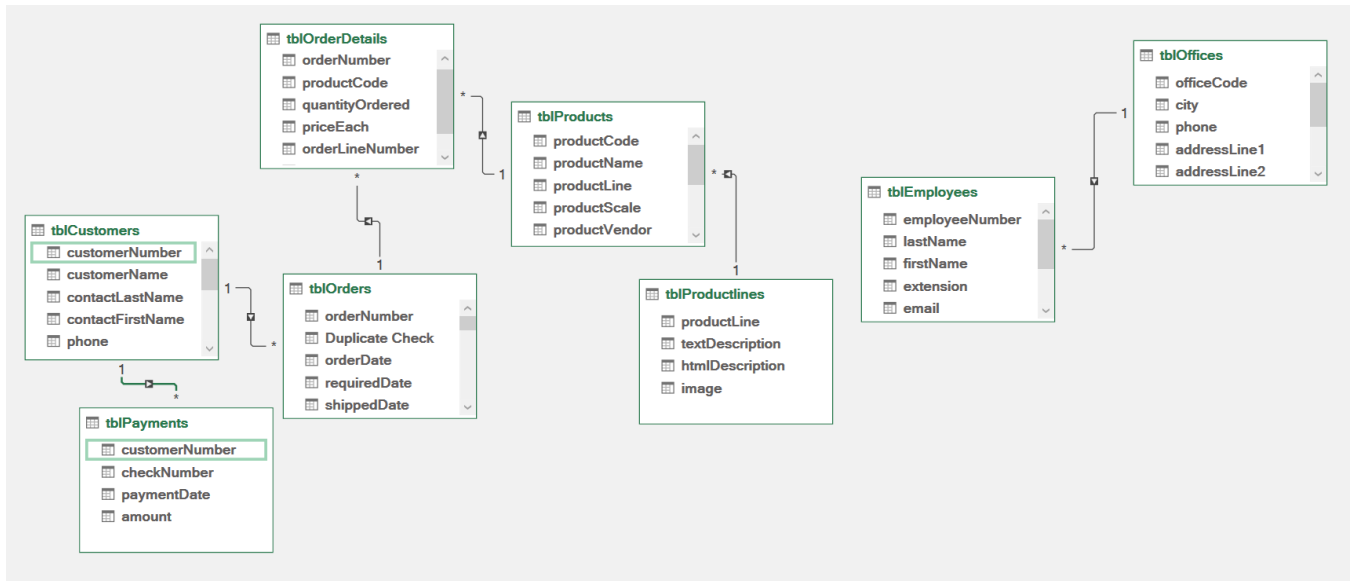
Relational model used to connect transactional, customer, product and organisational data

CONTROL PRINCIPLE

The model preserves business meaning. Transactional KPIs are filter-responsive where a valid relationship path exists; catalogue or snapshot measures remain static when the source does not contain the historical state required for a defensible period comparison.

3. Performance review

Revenue and order momentum



Revenue and order trends by year, quarter and month

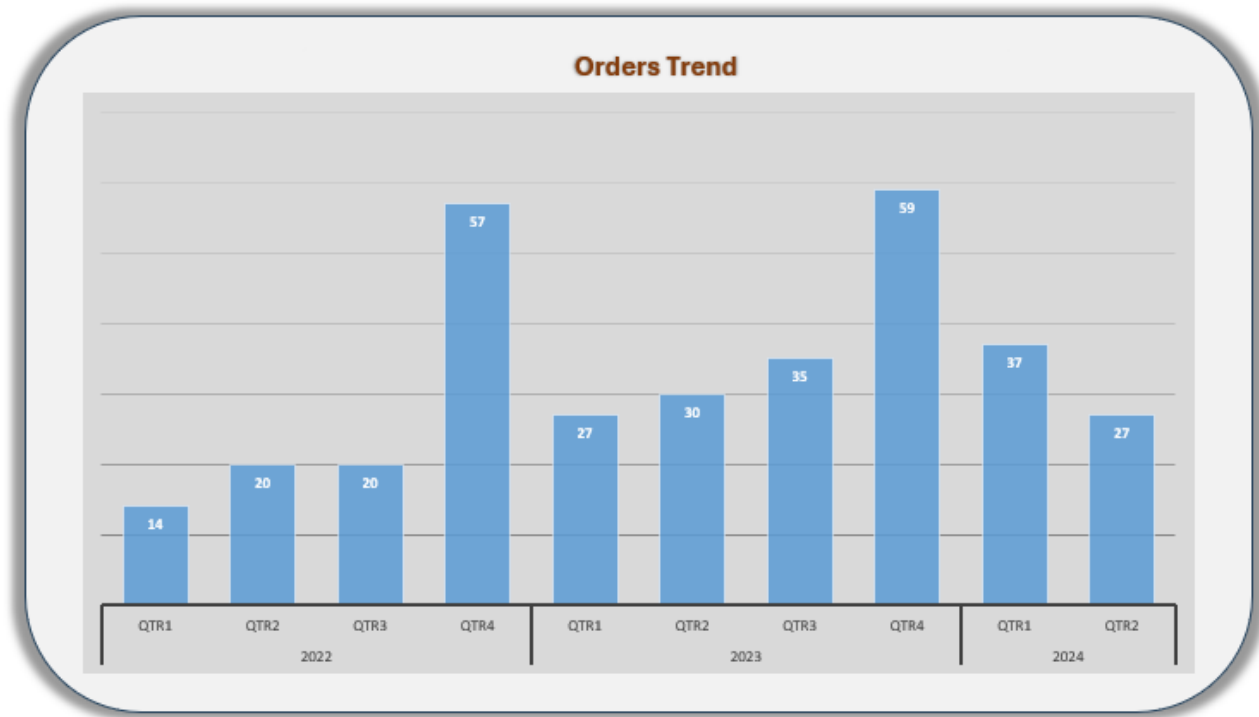
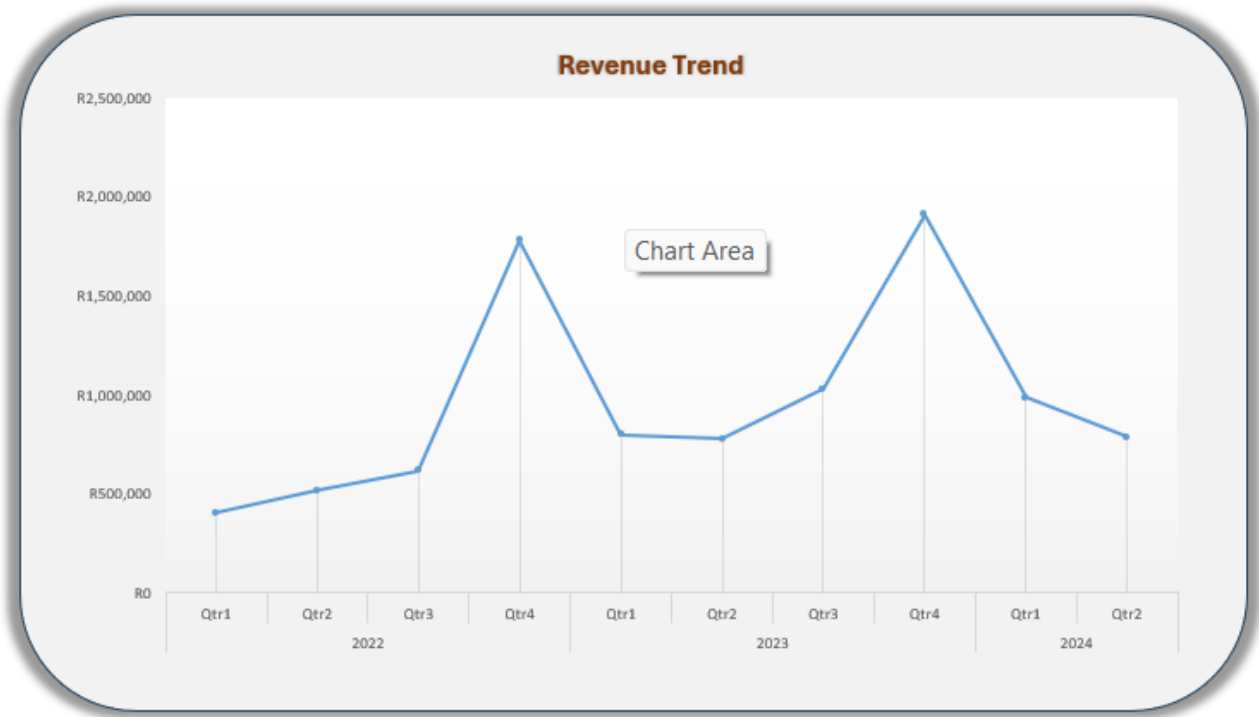
Revenue increased from approximately R3.32m in 2022 to R4.52m in 2023, an increase of about 36% across the two complete years. Order volume rose from 111 to 151 over the same period. Average order value remained broadly stable at approximately R29.9k, indicating that the improvement in revenue was driven primarily by greater transaction volume rather than a step-change in order size.

The most material pattern is the recurring Q4 peak. Q4 generated approximately R1.78m in 2022 and R1.91m in 2023, with 57 and 59 orders respectively. The repetition across two complete years makes seasonality a more decision-relevant observation than any single monthly movement. The January–May 2024 result of approximately R1.77m is partial-year and should be read as year-to-date performance only.

MANAGEMENT IMPLICATION

Treat Q3 as the readiness window for Q4. Inventory availability, supplier capacity, fulfilment staffing, marketing activity and cash-flow planning should be reviewed before the demand peak rather than after it becomes visible in the order book.

Product mix and concentration



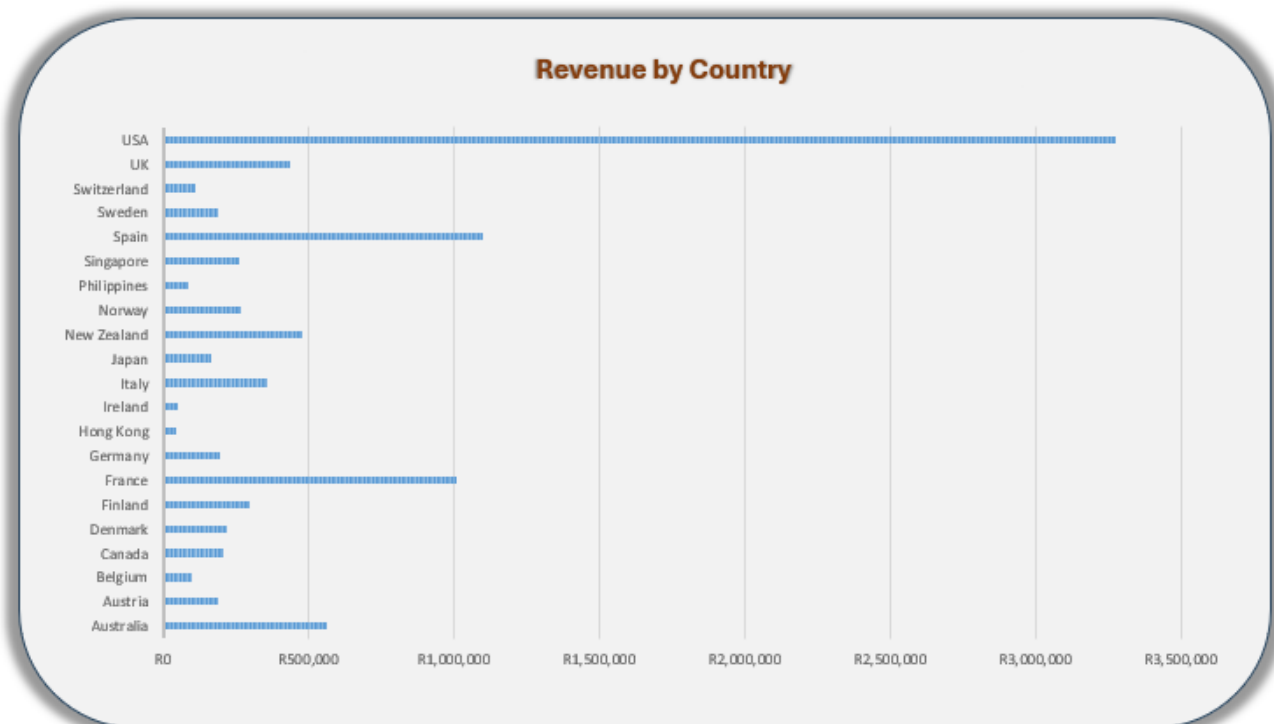
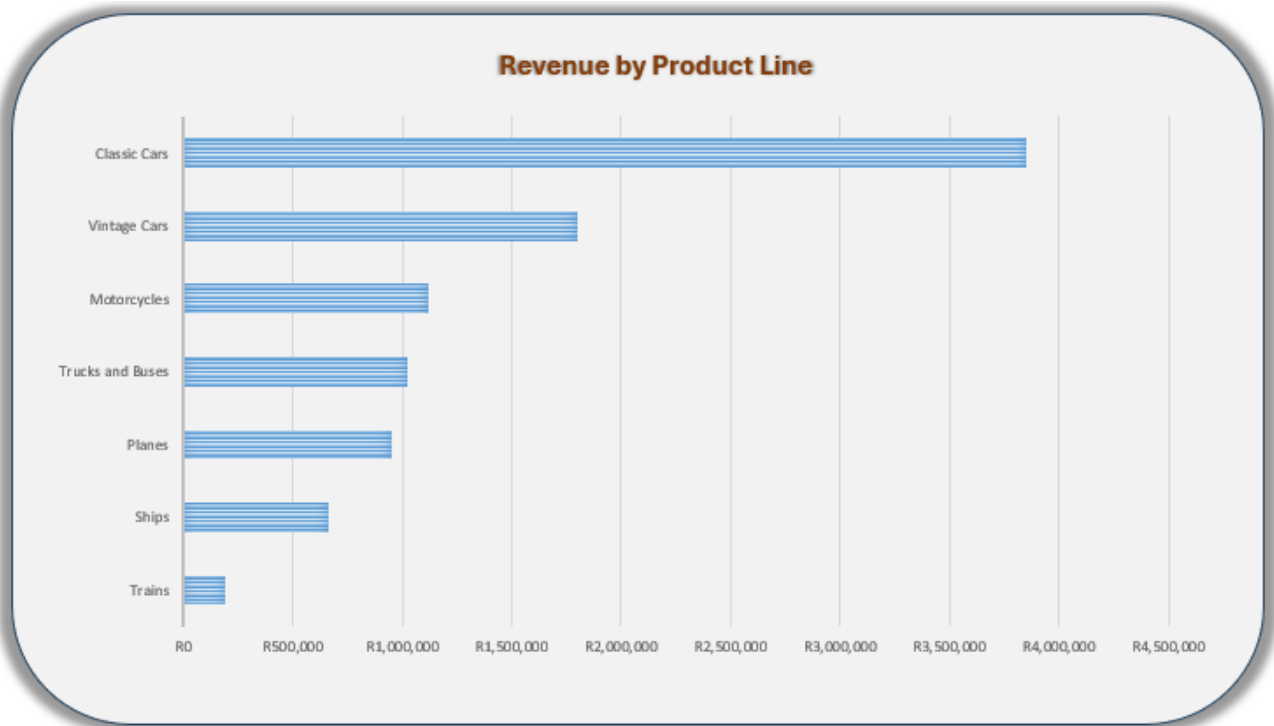
Revenue contribution by product line

Classic Cars generated approximately R3.85m, equivalent to about 40.1% of total revenue. Vintage Cars contributed approximately R1.80m, followed by Motorcycles (R1.12m), Trucks and Buses (R1.02m), Planes (R955k), Ships (R664k) and Trains (R189k). The portfolio therefore has a clear lead category, with a material gap between Classic Cars and the remainder.

MANAGEMENT IMPLICATION

Protect availability, pricing discipline and customer retention in Classic Cars, while testing whether adjacent categories can absorb additional commercial investment. Low revenue alone is not sufficient evidence to discontinue a category; margin, strategic role and cross-sell contribution would be required for that decision.

Geographic revenue profile



Revenue by country

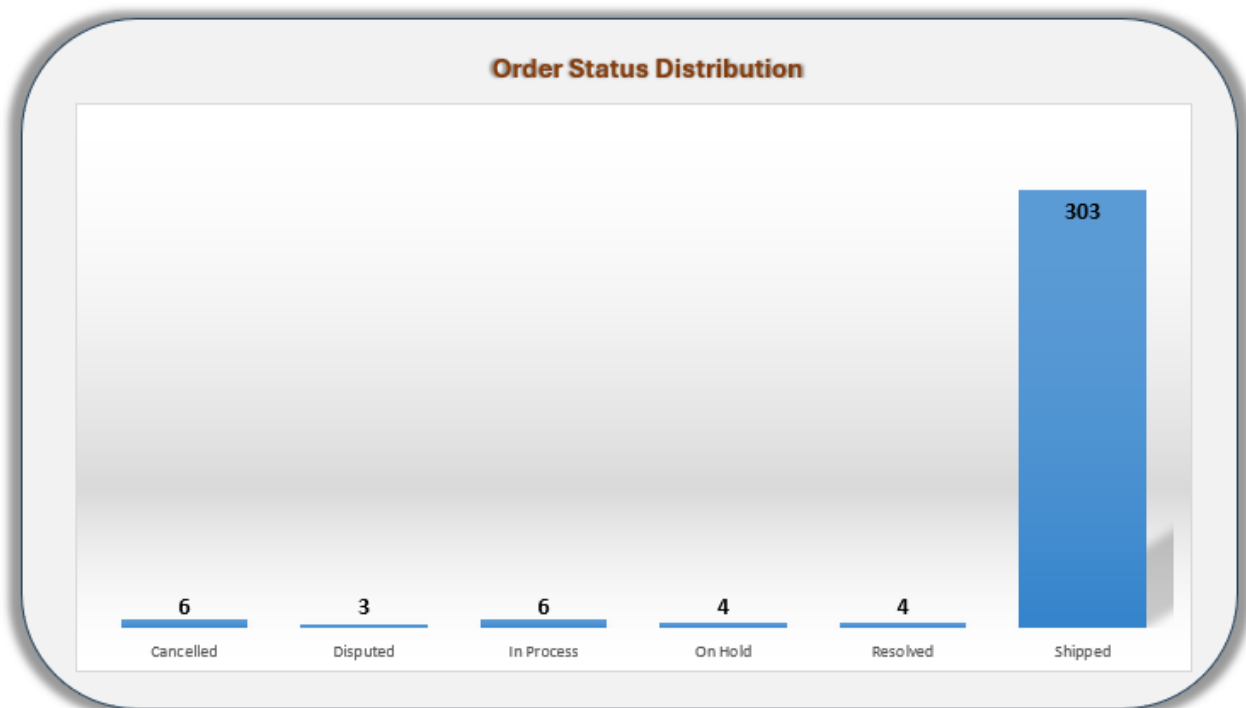
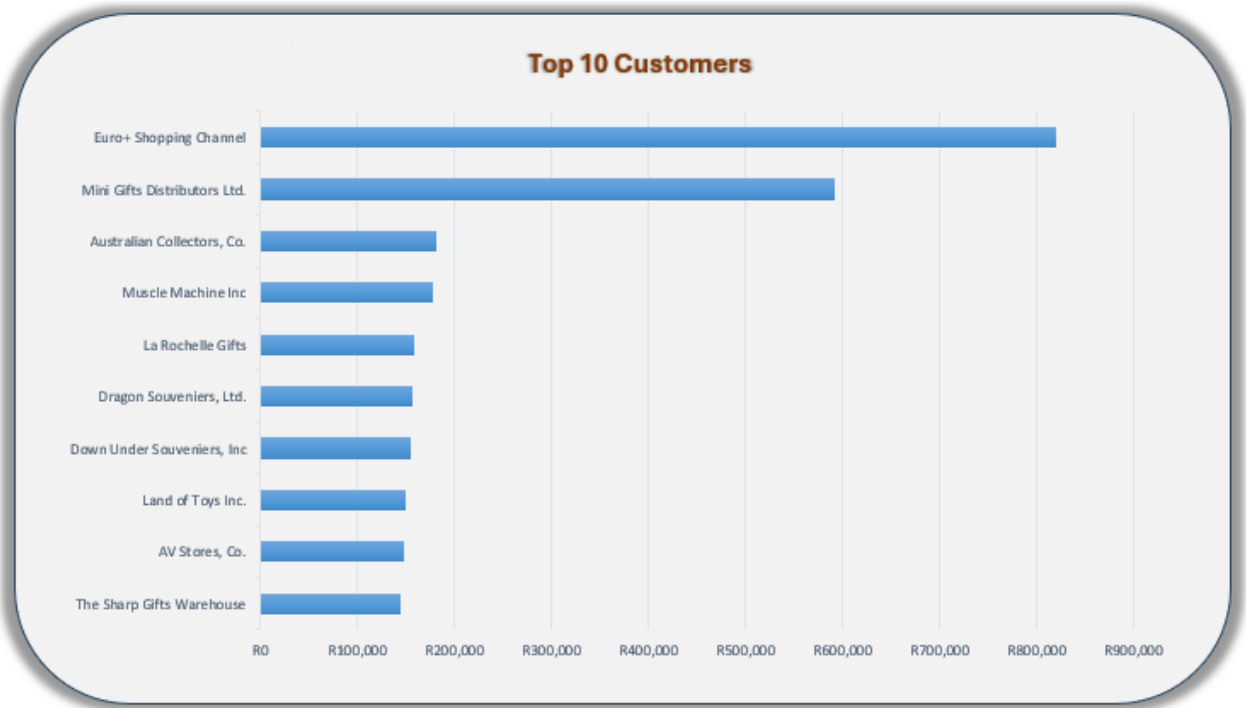
The United States generated approximately R3.27m, around 34% of total revenue. Spain (R1.10m) and France (R1.01m) are the next material markets, followed by Australia, New Zealand, the United Kingdom and Italy. The footprint is international, but value remains concentrated in a small number of markets.

MANAGEMENT IMPLICATION

The priority is not diversification for its own sake. Management should protect the economics of the US franchise while assessing whether the commercial playbook in the strongest markets can be replicated selectively in Spain, France and other markets with sufficient scale.

4. Customer and operational review

Customer concentration



Top 10 customers by revenue

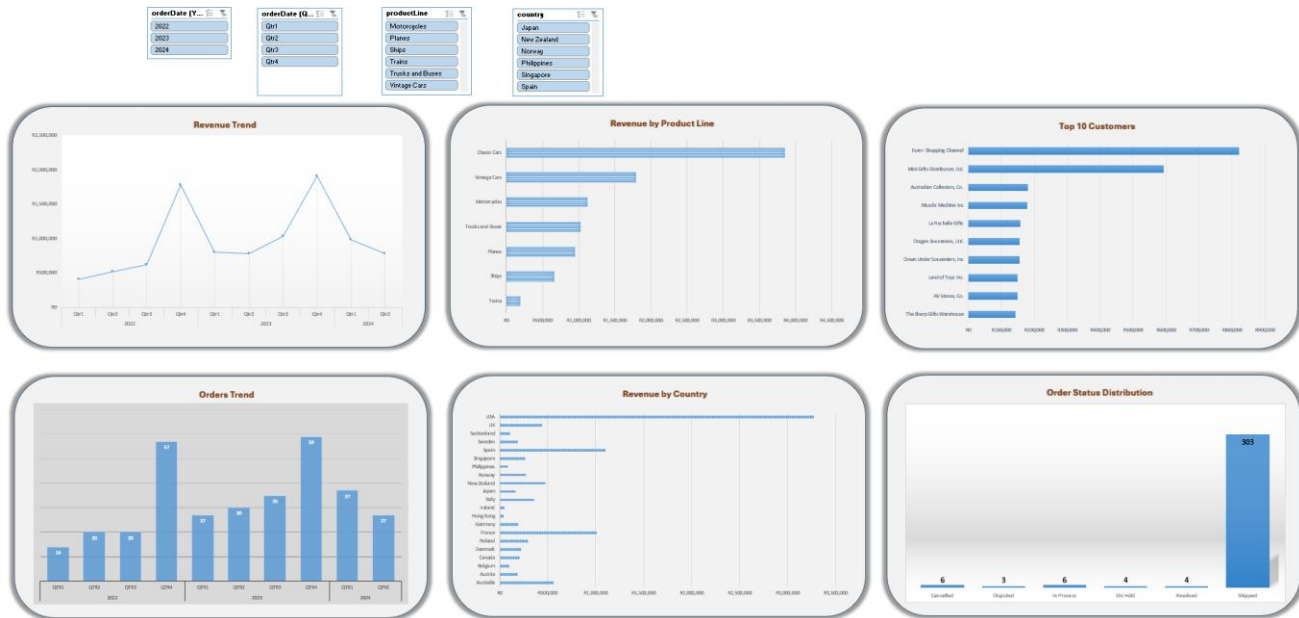
The top 10 customers generated approximately R2.68m, or 27.9% of total revenue. Euro+ Shopping Channel contributed approximately R821k and Mini Gifts Distributors Ltd. approximately R592k. The concentration is

material enough to warrant active account management, but the business is not dependent on a single customer for the majority of revenue.

MANAGEMENT IMPLICATION

Introduce a tiered key-account framework for the highest-value customers, combining retention monitoring, purchasing-frequency analysis, cross-sell opportunities and early-warning indicators for declining activity.

Order fulfilment and exceptions



Order status distribution

303 of 326 orders were shipped, producing a shipped-order rate of approximately 92.9%. Six orders were cancelled, six remained in process, four were on hold, four were resolved and three were disputed. The exception population is small enough to be reviewed individually rather than managed only through aggregate percentages.

MANAGEMENT IMPLICATION

Create a recurring exception review for cancelled, disputed and on-hold orders. Root-cause coding by customer, product, market and responsible sales representative would allow management to identify repeatable operational issues and track corrective action.

5. Interactive decision support

The dashboard uses slicers to move from an enterprise-wide view to a selected analytical context. Relevant transactional KPIs and charts recalculate by period, geography and product line, allowing management to test the drivers behind headline performance rather than relying on a static reporting pack.

The interactivity is deliberately governed by the data model. Measures such as current inventory and catalogue MSRP are not forced to respond to historical period selections where the source does not contain historical inventory or price snapshots. This design choice preserves the integrity of the management information.

SCREENSHOT PLACEMENT

Insert the filtered dashboard screenshot here — preferably a meaningful selection such as 2023 | Q4 | USA | Classic Cars — to demonstrate how the KPI layer and analytical charts respond to executive filters.

6. Strategic priorities arising from the analysis

1 | Q4 readiness

Formalise a Q3 readiness cycle covering inventory, supplier capacity, staffing, fulfilment capacity, marketing activation and cash-flow planning. The objective is to enter the recurring Q4 peak with capacity already aligned to expected demand.

2 | Product portfolio resilience

Protect the Classic Cars franchise, which contributes about 40% of revenue, while developing evidence on margin, demand and customer overlap across adjacent categories. The strategic objective is to retain the lead category without allowing concentration to become dependency.

3 | Market development

Maintain focus on the United States while testing targeted growth opportunities in Spain, France and other markets with sufficient revenue density. Expansion decisions should be based on customer economics and repeatability, not geographic presence alone.

4 | Key-account discipline

Segment the highest-value customers and establish explicit account ownership, retention triggers and cross-sell plans. Top-customer concentration is manageable, but the value at stake justifies structured relationship management.

5 | Operational exception management

Move cancelled, disputed and on-hold orders into a recurring exception-management process. Track root cause, owner, ageing and resolution to convert order-status reporting into an operational improvement mechanism.

6 | Management reporting cadence

Use the dashboard as a recurring management review tool rather than a one-off analytical output. A monthly cadence should focus on movement in the material KPIs, changes in known trends, emerging exceptions and the status of agreed management actions.

7. Solution capability and technical evidence

The project demonstrates the ability to move from raw multi-table data to an executive decision-support product. Technical work is presented as supporting evidence rather than as the main narrative of the case study.

Capability	Evidence in the solution
Data preparation	Structured Excel tables, data-type review, validation and revenue derivation
Data modelling	Relational model across orders, order lines, customers, products, payments, employees and offices
Analytical logic	DAX measures and KPI engines designed around valid filter context
Reporting	PivotTables, PivotCharts, time hierarchy, KPI cards and slicer-driven analysis
Business analysis	Trend, seasonality, concentration, market, customer and fulfilment analysis
Executive communication	Material findings linked to management implications and prioritised actions

Tools and methods

Microsoft Excel • Excel Tables • Power Pivot • DAX • Relational Data Modelling • PivotTables • PivotCharts • Slicers • KPI Design • Data Validation • Time-Series Analysis • Executive Reporting

8. Conclusion

The Executive Business Intelligence Suite converts fragmented operational records into a connected management view of performance. Its value lies not in the number of charts produced, but in the discipline of linking measures to business questions, distinguishing material signals from supporting detail and translating observed patterns into a focused management agenda.

The resulting case study demonstrates both sides of the Business Operations & Data Analytics proposition: the technical ability to structure, model and interrogate business data, and the commercial ability to explain what the information means for management.